HILBERT 90 AND KUMMER THEORY

SANTIAGO ARANGO-PIÑEROS

Hilbert's Theorem 90 says that $H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$. Applied to the Kummer sequence $1 \to \mu_m \to \mathbb{G}_{\mathrm{m}} \xrightarrow{m} \mathbb{G}_{\mathrm{m}} \to 1$, it gives an isomorphism $k^\times/k^{\times m} \cong H^1(k, \mu_m)$ with an explicit formula, and, when k contains the m-th roots of unity, a classification of the abelian extensions of k of exponent dividing m. The problems below prove all of this from the definitions in [Sil09, Appendix B]. Problem 1.3 computes cohomology groups by hand, Problem 2.5 proves Hilbert 90, Problem 3.4 unwinds the Kummer sequence and Problem 4.2 derives Kummer theory. Problems 5.4 and 5.7 compute the groups $k^\times/k^{\times m}$ for finite, local and global fields, and cut out a finite subgroup by conditions at the primes. Later, the same sequence for an elliptic curve E will give an injection $E(k)/m \hookrightarrow H^1(k, E[m])$. Then there is no Hilbert 90, and its failure is measured by $H^1(k, E)$. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A field k, not assumed perfect, a separable closure k^{sep} of k, and the absolute Galois group $\mathrm{Gal}_k := \mathrm{Gal}(k^{\mathrm{sep}}/k)$. For an extension L of k inside k^{sep}, $\mathrm{Gal}_L := \mathrm{Gal}(k^{\mathrm{sep}}/L)$. (Silverman's Appendix B assumes k perfect and works with an algebraic closure. Nothing in it changes with k^{sep} in its place.)
- A profinite group G, often finite, and a discrete G-module A: an abelian group with a left action of G, written $a \mapsto ga$, such that $G \times A \to A$ is continuous for the discrete topology on A. (Silverman writes right actions, $a \mapsto a^\sigma$.)
- Elements g, h of G, and elements σ, τ, γ of a Galois group such as Gal_k or $\mathrm{Gal}(L/k)$. Where σ is a generator of a cyclic group, it keeps that meaning.
- An integer $m \geqslant 1$ with $\mathrm{char}\, k \nmid m$, and the algebraic group μ_m of m-th roots of unity, the kernel of $x \mapsto x^m$ on $\mathbb{G}_{\mathrm{m}}$. For a field L containing k, $\mu_m(L)$ is the group of m-th roots of unity in L.
- **An abuse of notation.** For a commutative algebraic group A over k, such as $\mathbb{G}_{\mathrm{m}}$, μ_m or an elliptic curve, we write $H^i(k, A) := H^i(\mathrm{Gal}_k, A(k^{\mathrm{sep}}))$, the cohomology of the Gal_k-module of k^{sep}-points of A. For instance,

$$H^1(k, \mathbb{G}_{\mathrm{m}}) = H^1(\mathrm{Gal}_k, (k^{\mathrm{sep}})^\times), \qquad H^1(k, \mu_m) = H^1(\mathrm{Gal}_k, \mu_m(k^{\mathrm{sep}})).$$

  This is the only place where A stands for $A(k^{\mathrm{sep}})$.
- The quotient $B/m := B/mB$ of an abelian group B. For the multiplicative group of a field L it is $L^\times/L^{\times m}$, where $L^{\times m}$ is the subgroup of m-th powers.

How to use this. Allow about 15 minutes for Problem 1.3, 25 for Problem 2.5 and 25 for Problem 3.4. The heart of Problem 2.5 is items (1) and (4), and that of Problem 3.4 is items (2) and (3). If time is short, leave Problem 2.5(5) and Problem 3.4(5) for later, like every item marked $\star$. Problems 4.2, 5.4 and 5.7 are for home. The `Magma` blocks run in seconds in the free online calculator. Everything else is by hand.

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1. Cocycles by hand

This section uses the definitions of [Sil09, §§B.1–B.2] and [Poo02, §§2–3]. Its last item is [Sil09, Exercise B.2(a)].

Definition 1.1 (Cohomology in degrees 0 and 1 [Sil09, §§B.1–B.2], [Poo02, §3])**.** Let A be a discrete G-module. Then $H^0(G,A) \coloneqq A^G$, the subgroup of the $a \in A$ with $ga = a$ for every $g \in G$. A left 1-cocycle, or simply 1-cocycle, is a continuous map $\xi\colon G \to A$, $g \mapsto \xi_g$, such that

$$\xi_{gh} = \xi_g + g\xi_h \qquad \text{for all } g,h \in G.$$

A left 1-coboundary, or simply 1-coboundary, is a map of the form $g \mapsto ga - a$, for some $a \in A$. Both are *left*: they are written for the left action $a \mapsto ga$. Silverman's are the right ones, for $a \mapsto a^\sigma$: $\xi_{\sigma\tau} = \xi_\sigma^\tau + \xi_\tau$ and $\xi_\sigma = a^\sigma - a$. The 1-cocycles form a group $Z^1(G,A)$ under addition of values, the 1-coboundaries a subgroup $B^1(G,A)$, and $H^1(G,A) \coloneqq Z^1(G,A)/B^1(G,A)$. If G is finite, every map $G \to A$ is continuous. When the group law of A is written multiplicatively, as for $L^\times$ or $\mu_m(L)$, the conditions read $\xi_{gh} = \xi_g \cdot g(\xi_h)$ and $\xi_g = g(a)/a$.

Proposition 1.2 (Trivial action [Sil09, Remarks B.1.1 and B.2.2], [Poo02, §3])**.** *If G acts trivially on A, then $H^0(G,A) = A$ and $H^1(G,A) = \mathrm{Hom}_{\mathrm{conts}}(G,A)$, the group of continuous homomorphisms.*

Problem 1.3 (Computing H^1 from the definition)**.**
In this problem G is finite.
(1) Let $G = \{1,\sigma\}$ have order 2, acting on $\mathbb{Z}$ by $\sigma a = -a$. Compute $H^0(G,\mathbb{Z})$ and $H^1(G,\mathbb{Z})$. Compare with the trivial action of G on $\mathbb{Z}$.
(2) Let $G = \langle\sigma\rangle$ be cyclic of order n, and $N \coloneqq 1 + \sigma + \cdots + \sigma^{n-1}$, acting on A. Show that $\xi \mapsto \xi_\sigma$ induces an isomorphism[a]

$$H^1(G,A) \cong \ker(N\colon A \to A)/(\sigma - 1)A.$$

(3) Let $G = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1,c\}$. Use (2) to compute $H^1(G,\mathbb{C}^\times)$ and $H^1(G,\mu_m(\mathbb{C}))$. Compare the order of the second group with that of $\mathbb{R}^\times/\mathbb{R}^{\times m}$.
(4) Suppose that G has order n. Show that $nH^1(G,A) = 0$. [b]

[a] *Hint:* show that $\xi_{\sigma^i} = (1 + \sigma + \cdots + \sigma^{i-1})\xi_\sigma$ for $i \geqslant 1$.
[b] *Hint:* sum the cocycle condition $\xi_{gh} = \xi_g + g\xi_h$ over $h \in G$.

2. Hilbert 90

This section is [Poo17, Proposition 1.3.15(ii) and Exercises 1.10–1.11], with the proof of [Mil20, Proposition IV.1.3] (also [Mil22, Theorem 5.23]). [Sil09, Proposition B.2.5] states the result and refers to Serre for the proof. Item (2) is the form in which Hilbert stated it.

Theorem 2.1 (Dedekind [Mil22, Theorem 5.14])**.** *Let F be a field and H a group. If $\chi_1,\dots,\chi_r$ are distinct homomorphisms $H \to F^\times$, then they are linearly independent over F: if $a_1\chi_1 + \cdots + a_r\chi_r = 0$ as a function $H \to F$, with $a_i \in F$, then $a_1 = \cdots = a_r = 0$.*

Definition 2.2 (Norm and trace)**.** Let L/k be a finite Galois extension with group G. The norm and the trace of $a \in L$ are $N_{L/k}(a) \coloneqq \prod_{\sigma\in G}\sigma(a)$ and $\mathrm{Tr}_{L/k}(a) \coloneqq \sum_{\sigma\in G}\sigma(a)$. Both lie in k.

Proposition 2.3 (Infinite Galois theory)**.**

(1) The subgroups Gal_L, for L running over the finite Galois extensions of k inside k^{sep}, are open and normal in Gal_k, and they form a basis of open neighborhoods of 1 [Sil09, §B.2].

(2) The maps $H \mapsto (k^{\mathrm{sep}})^H$ and $L \mapsto \mathrm{Gal}_L$ are inverse bijections between the closed subgroups of Gal_k and the extensions of k inside k^{sep}. The subgroup H is open if and only if $(k^{\mathrm{sep}})^H$ is finite over k, and then $[\mathrm{Gal}_k : H] = [(k^{\mathrm{sep}})^H : k]$. It is normal if and only if $(k^{\mathrm{sep}})^H$ is Galois over k, and then $\mathrm{Gal}((k^{\mathrm{sep}})^H/k) \cong \mathrm{Gal}_k/H$ by restriction. In particular, $(k^{\mathrm{sep}})^{\mathrm{Gal}_k} = k$ [Mil22, Theorem 7.13].

Definition 2.4 (Restriction and inflation [Sil09, §B.2], [Poo02, §4])**.** Let A be a discrete G-module and $H \subseteq G$ a closed subgroup. The restriction $\mathrm{Res}\colon H^1(G, A) \to H^1(H, A)$ maps the class of ξ to the class of $\xi|_H$. If H is normal, the inflation $\mathrm{Inf}\colon H^1(G/H, A^H) \to H^1(G, A)$ maps the class of ξ to the class of the composition $G \to G/H \xrightarrow{\xi} A^H \subseteq A$. A 1-cocycle of G of this form is inflated from G/H.

Problem 2.5 (Hilbert's Theorem 90)**.**

(1) Let L/k be a finite Galois extension with group G, and $\xi\colon G \to L^\times$ a 1-cocycle. Show that there is a $c \in L^\times$ with
$$b := \sum_{\tau \in G} \xi_\tau \cdot \tau(c) \neq 0,$$
and that then $\xi_\sigma = b/\sigma(b)$ for every $\sigma \in G$. Deduce that $H^1(G, L^\times) = 1$. [a]
(2) Suppose moreover that $G = \langle\sigma\rangle$ is cyclic. Show that if $a \in L^\times$ has $N_{L/k}(a) = 1$, then $a = \sigma(b)/b$ for some $b \in L^\times$.
(3) Let (x, y) be a rational point of the circle $x^2 + y^2 = 1$. Apply (2) to $\mathbb{Q}(i)/\mathbb{Q}$ to find $u, v \in \mathbb{Q}$, not both 0, with
$$(x, y) = \Big(\frac{u^2 - v^2}{u^2 + v^2}, \frac{2uv}{u^2 + v^2}\Big).$$
(4) Let $\xi\colon \mathrm{Gal}_k \to (k^{\mathrm{sep}})^\times$ be a continuous 1-cocycle.
 (a) Show that there is a finite Galois extension L of k inside k^{sep} such that ξ takes values in $L^\times$ and $\xi_{\sigma\tau} = \xi_\sigma$ for all $\sigma \in \mathrm{Gal}_k$ and $\tau \in \mathrm{Gal}_L$. So ξ is inflated from a 1-cocycle $\mathrm{Gal}(L/k) \to L^\times$. [b]
 (b) Deduce that $H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$.
(5) ($\star$) Let L/k be finite Galois with group G, and $\xi\colon G \to L$ a 1-cocycle for the additive group of L. Show that $H^1(G, L) = 0$, by a variant of (1). [c]

[a] *Hint:* compute $\sigma(b)$ with the cocycle condition.
[b] *Hint:* Gal_k is compact, and $\xi^{-1}(1)$ is open.
[c] *Hint:* take $c \in L$ with $\mathrm{Tr}_{L/k}(c) \neq 0$.

3. The Kummer sequence

This section is the exercise in [Poo02, §3], with the explicit formula of [Sil09, Propositions B.2.5(c) and VIII.2.2]. Hilbert's Theorem 90, $H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$, is Problem 2.5(4).

Proposition 3.1 (Finite multiplicative groups [Mil22, Exercise 1-3])**.** *Every finite subgroup of the multiplicative group of a field is cyclic.*

Proposition 3.2 (The long exact sequence [Sil09, Propositions B.1.2 and B.2.3], [Poo02, Theorem 1])**.** *Let $0 \to A \to B \xrightarrow{\psi} C \to 0$ be an exact sequence of discrete G-modules. There is an exact sequence*
$$0 \to A^G \to B^G \to C^G \xrightarrow{\delta} H^1(G, A) \to H^1(G, B) \to H^1(G, C),$$
where, for $c \in C^G$, $\delta(c)$ is the class of $g \mapsto gb - b$ for any $b \in B$ with $\psi(b) = c$.

Proposition 3.3 (Inflation–restriction [Sil09, Proposition B.2.4]). *Let A be a discrete Gal_k-module and L/k a finite Galois extension inside k^{sep}. The sequence*

$$0 \to H^1(\mathrm{Gal}(L/k), A^{\mathrm{Gal}_L}) \xrightarrow{\mathrm{Inf}} H^1(\mathrm{Gal}_k, A) \xrightarrow{\mathrm{Res}} H^1(\mathrm{Gal}_L, A)$$

is exact.

Problem 3.4 (The Kummer sequence).
Recall that $\operatorname{char} k \nmid m$, so $m \neq 0$ in k.
(1) Show that $\mu_m(k^{\mathrm{sep}})$ is cyclic of order m, and that

$$1 \to \mu_m(k^{\mathrm{sep}}) \to (k^{\mathrm{sep}})^\times \xrightarrow{x \mapsto x^m} (k^{\mathrm{sep}})^\times \to 1$$

is an exact sequence of discrete Gal_k-modules. Deduce from Proposition 3.2 that its connecting map induces an isomorphism $\delta\colon k^\times/k^{\times m} \to H^1(k, \mu_m)$.
(2) Let $a \in k^\times$ and $\alpha \in k^{\mathrm{sep}}$ with $\alpha^m = a$. Check directly, without Proposition 3.2, that $\xi_\sigma := \sigma(\alpha)/\alpha$ is a continuous 1-cocycle with values in $\mu_m(k^{\mathrm{sep}})$, whose class does not depend on the choice of α. Show that $a \mapsto [\xi]$ is a homomorphism $k^\times \to H^1(k, \mu_m)$ with kernel $k^{\times m}$. This is the map δ of (1).
(3) Show directly from Hilbert 90 that every continuous 1-cocycle $\xi\colon \mathrm{Gal}_k \to \mu_m(k^{\mathrm{sep}})$ is cohomologous to one of the form of (2).
(4) Let $m = 2$, with $\operatorname{char} k \neq 2$. Show that $H^1(k, \mu_2) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \{\pm 1\})$, and that for $d \in k^\times - k^{\times 2}$, $\delta(d)$ is the homomorphism $\chi_d\colon \sigma \mapsto \sigma(\sqrt{d})/\sqrt{d}$, with kernel $\mathrm{Gal}_{k(\sqrt{d})}$. Deduce that $d \mapsto k(\sqrt{d})$ is a bijection from the nontrivial elements of $k^\times/k^{\times 2}$ to the quadratic extensions of k inside k^{sep}.
(5) Let L be an extension of k inside k^{sep}. Show that the restriction $H^1(k, \mu_m) \to H^1(L, \mu_m)$ corresponds, under δ, to the map $k^\times/k^{\times m} \to L^\times/L^{\times m}$ induced by inclusion, and that its kernel corresponds to $(k^\times \cap L^{\times m})/k^{\times m}$. Compute the kernel for $\mathbb{Q}(i)/\mathbb{Q}$ and $m = 2$, and compare it with Proposition 3.3.

4. Kummer theory

This section is [Sil09, Exercise 8.4], in the form of [Mil22, Theorem 5.30 and Remark 5.32], together with an example that shows why the roots of unity are needed. In it, $\mu_m(k^{\mathrm{sep}}) \subseteq k$, except in item (4).

Proposition 4.1 (Characters of a finite abelian group [Mil22, discussion before Theorem 5.30]). *If B is a finite abelian group of exponent dividing m, then $\mathrm{Hom}(B, \mu_m(k^{\mathrm{sep}}))$ has the same order as B.*

Problem 4.2 (Abelian extensions of exponent dividing m).

(1) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Show that δ is an isomorphism

$$\delta\colon k^\times/k^{\times m} \to \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \mu_m(k^{\mathrm{sep}})).$$

For $a \in k^\times$ and $\alpha^m = a$, show that $\ker \delta(a) = \mathrm{Gal}_{k(\alpha)}$, and deduce that $k(\alpha)/k$ is Galois and cyclic, of degree the order of a in $k^\times/k^{\times m}$.
(2) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Let Δ be a subgroup of $k^\times$ containing $k^{\times m}$, with $\Delta/k^{\times m}$ finite, and $L := k(\Delta^{1/m})$ the extension of k inside k^{sep} generated by the m-th roots of the elements of

Δ. Show that L/k is finite and Galois, and that

$$\langle\ ,\ \rangle\colon \mathrm{Gal}(L/k) \times \Delta/k^{\times m} \to \mu_m(k^{\mathrm{sep}}), \qquad \langle \sigma, a\rangle \coloneqq \sigma(\alpha)/\alpha \quad (\alpha^m = a),$$

is a well defined bilinear pairing, and that it is perfect: the induced maps

$$\mathrm{Gal}(L/k) \to \mathrm{Hom}(\Delta/k^{\times m}, \mu_m(k^{\mathrm{sep}})), \qquad \Delta/k^{\times m} \to \mathrm{Hom}(\mathrm{Gal}(L/k), \mu_m(k^{\mathrm{sep}}))$$

are isomorphisms. Deduce that $\mathrm{Gal}(L/k)$ is abelian of exponent dividing m and that $[L:k] = (\Delta : k^{\times m})$.[a]

(3) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Show that $\Delta \mapsto k(\Delta^{1/m})$ is a bijection from the subgroups Δ as in (2) to the finite abelian extensions of k inside k^{sep} of exponent dividing m, with inverse $L \mapsto \Delta_L \coloneqq k^\times \cap L^{\times m}$.[b]

(4) ($\star$) Let $k = \mathbb{Q}$, $m = 3$, $\zeta \in \overline{\mathbb{Q}}$ a primitive cube root of unity, and $F \coloneqq \mathbb{Q}(\zeta)$. Here $\mu_3(\overline{\mathbb{Q}}) \not\subseteq \mathbb{Q}$.

(a) Show that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not Galois, although $2 \notin \mathbb{Q}^{\times 3}$. Which step of (1) fails?

(b) Show that $\mathbb{Q}^\times/\mathbb{Q}^{\times 3} \to F^\times/F^{\times 3}$ is injective.[c]

(c) Let $a \in \mathbb{Q}^\times - \mathbb{Q}^{\times 3}$, $\alpha^3 = a$, and $L \coloneqq F(\alpha)$. Show that $L/\mathbb{Q}$ is Galois of degree 6, and that conjugation by any $\tau \in \mathrm{Gal}(L/\mathbb{Q})$ with $\tau(\zeta) = \zeta^{-1}$ acts on $\mathrm{Gal}(L/F) \cong \mu_3(F)$, the isomorphism of (1) over F, as inversion. Deduce that $\mathrm{Gal}(L/\mathbb{Q}) \cong S_3$. Check with `Magma`:

```
R<x> := PolynomialRing(Rationals());
G := GaloisGroup(x^3 - 2);
#G, IsAbelian(G);                    // 6 false
F<zeta> := CyclotomicField(3);
S<y> := PolynomialRing(F);
#GaloisGroup(y^3 - 2);               // 3
```

[a] *Hint:* show that both maps are injective, then count with Proposition 4.1.

[b] *Hint:* for L given, compute $\Delta_L/k^{\times m}$ with Problem 3.4(5) and Proposition 3.3.

[c] *Hint:* Proposition 3.3 and Problem 1.3(4).

5. Computing $k^\times/k^{\times m}$

This section computes $k^\times/k^{\times m} \cong H^1(k, \mu_m)$ for finite fields, $\mathbb{Q}_p$ and $\mathbb{Q}$, and then cuts out a finite subgroup by conditions at the primes, as in [Poo02, §§9–10]. The last item is [Sil09, proof of Proposition VIII.1.6], with a `Magma` check.

Lemma 5.1 (Hensel's lemma [Sil09, Lemma IV.1.2])**.** *Let $F \in \mathbb{Z}_p[w]$ and $a \in \mathbb{Z}_p$ with $F(a) \in p^n\mathbb{Z}_p$ for some $n \geqslant 1$ and $F'(a) \in \mathbb{Z}_p^\times$. Then F has a root $b \in \mathbb{Z}_p$ with $b \equiv a \pmod{p^n}$.*

Definition 5.2 (Unramified classes [Poo02, §§4 and 9])**.** Let p be a prime. Embed $\overline{\mathbb{Q}}$ in an algebraic closure $\overline{\mathbb{Q}}_p$ of $\mathbb{Q}_p$, so that restriction is an injection $\mathrm{Gal}_{\mathbb{Q}_p} \coloneqq \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \hookrightarrow \mathrm{Gal}_{\mathbb{Q}}$. Let ${\mathbb{Q}_p}^{\mathrm{unr}} \subseteq \overline{\mathbb{Q}}_p$ be the maximal unramified extension of $\mathbb{Q}_p$, the union of its finite unramified extensions. A finite extension of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}}_p$ is unramified if and only if it lies in ${\mathbb{Q}_p}^{\mathrm{unr}}$. The inertia group is $I_p \coloneqq \mathrm{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\mathrm{unr}}) \subseteq \mathrm{Gal}_{\mathbb{Q}_p}$. A class $\xi \in H^1(\mathbb{Q}, A)$, for a commutative algebraic group A over $\mathbb{Q}$, is unramified at p if its image in $H^1(I_p, A(\overline{\mathbb{Q}}_p))$ is trivial. For a finite set S of primes, $H^1(\mathbb{Q}, A; S)$ is the subgroup of the classes unramified at every prime $p \notin S$.

Proposition 5.3 (Unramified Kummer extensions [Sil09, proof of Proposition VIII.1.6])**.** *Let p be a prime with $p \nmid m$, $a \in {\mathbb{Q}_p}^\times$ and $\alpha \in \overline{\mathbb{Q}}_p$ with $\alpha^m = a$. Then $\mathbb{Q}_p(\alpha)/\mathbb{Q}_p$ is unramified if and only if $v_p(a) \equiv 0 \pmod{m}$.*

Problem 5.4 (Finite and local fields)**.**

(1) Let q be a power of a prime not dividing m. Show that $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ is cyclic of order $\gcd(m, q-1)$. If $m \mid q-1$, so that $\mu_m(\overline{\mathbb{F}}_q) \subseteq \mathbb{F}_q$, compare the subgroups of $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ with the extensions of $\mathbb{F}_q$ of degree dividing m, as Problem 4.2(3) predicts.
(2) Let p be odd and $u \in \mathbb{Z}_p^\times$ a unit whose reduction is not a square in $\mathbb{F}_p$. Show that ${\mathbb{Q}_p}^\times/{\mathbb{Q}_p}^{\times 2}$ has order 4, with representatives $1, u, p, up$. List the quadratic extensions of $\mathbb{Q}_p$, and say which one is unramified.[a]
(3) Show that a unit $v \in \mathbb{Z}_2^\times$ is a square if and only if $v \equiv 1 \pmod 8$. Deduce that $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$ has order 8, with representatives $\pm 1, \pm 2, \pm 5, \pm 10$. `Magma` counts the quadratic extensions of $\mathbb{Q}_p$:

```
#AllExtensions(pAdicRing(3, 20), 2);    // 3
#AllExtensions(pAdicRing(5, 20), 2);    // 3
#AllExtensions(pAdicRing(2, 20), 2);    // 7
```

Check these against your answers.[b]

[a] *Hint:* every element of ${\mathbb{Q}_p}^\times$ is $p^n v$ with $n \in \mathbb{Z}$ and $v \in \mathbb{Z}_p^\times$, and Lemma 5.1 decides which units are squares.
[b] *Hint:* for $v \equiv 1 \pmod 8$, apply Lemma 5.1 to $F(w) = w^2 + w - (v-1)/4$, then look at $(2w+1)^2$.

Definition 5.5 (The group $k(S, m)$ [Sil09, Theorem X.1.1(c)])**.** Let k be a number field, S a finite set of places of k containing the archimedean ones, and

$$k(S, m) := \{a \in k^\times/k^{\times m} : v(a) \equiv 0 \pmod m \text{ for all } v \notin S\},$$

where v runs over the normalized valuations of k. Let $\mathcal{O}_S$ be the ring of S-integers, the $a \in k$ with $v(a) \geqslant 0$ for all $v \notin S$, and $\mathrm{Cl}(\mathcal{O}_S)$ its ideal class group. For $k = \mathbb{Q}$ we list only the primes in S, and in [Poo02, §10], $\mathbb{Q}(S, 2)$ is written $\langle -1, S\rangle$.

Proposition 5.6 (Finiteness [Sil09, proof of Proposition VIII.1.6])**.** *The group* $\mathrm{Cl}(\mathcal{O}_S)$ *is finite, and* $\mathcal{O}_S^\times$ *is finitely generated (Dirichlet's S-unit theorem).*

Problem 5.7 (Classes unramified outside S)**.**

(1) Show that $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ is an $\mathbb{F}_2$-vector space with basis the classes of -1 and of the primes. In particular, it is infinite.
(2) Let S be a finite set of primes. Show that $\mathbb{Q}(S, 2)$ is the subgroup generated by -1 and the primes in S, of order $2^{\#S+1}$.
(3) Let p be an odd prime and $a \in \mathbb{Q}^\times$. Show that $\delta(a) \in H^1(\mathbb{Q}, \mu_2)$ is unramified at p if and only if $v_p(a)$ is even. Deduce that $\mathbb{Q}(S, 2) = \delta^{-1}H^1(\mathbb{Q}, \mu_2; S)$ for every finite set S of primes containing 2.
(4) ($\star$) Let k be a number field and S as in Definition 5.5. Construct an exact sequence[a]

$$1 \to \mathcal{O}_S^\times/\mathcal{O}_S^{\times m} \to k(S, m) \to \mathrm{Cl}(\mathcal{O}_S)[m] \to 0,$$

and deduce that $k(S, m)$ is finite. For $k = \mathbb{Q}(\sqrt{-5})$, $m = 2$ and S the archimedean place, show that $k(S, 2) = \langle -1, 2\rangle$, of order 4, and check with `Magma`:

```
K<s> := QuadraticField(-5);
OK := Integers(K);
#ClassGroup(OK);                    // 2
```

```
G, m := pSelmerGroup(2, {PowerIdeal(OK)|});
#G;                                    // 4
[K | G.i @@ m : i in [1..Ngens(G)]];  // [ -1, 2 ]
```

[a] *Hint:* the prime ideals of $\mathcal{O}_S$ are those of the places outside S. If a represents an element of $k(S,m)$, then $a\mathcal{O}_S = \mathfrak{a}^m$ for a fractional ideal $\mathfrak{a}$ of $\mathcal{O}_S$.

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